1. Which cloud-based accounting information system should Pia select and why?

Through the case information, Pia runs a small pharmacy with a complex staff structure and needs to process a large number of invoices and payroll. She currently uses Excel and some professional software, but the proportion of electronic invoicing is low. She needs to improve efficiency and reduce administrative burden while ensuring data security and compliance.

System 3 seems to be the best option for Pia due to the following reasons:

1). Predictable Pricing.

System 3 has a monthly fee of 150 euros with no transaction fee and 2 hours of accounting services (cost 120 euros). This pricing model makes it easier for Pia to budget her accounting costs and avoids the unpredictable transaction fees as her business grows.

System 1 and System 2 have lower initial monthly fees (20 euros and 69 euros, respectively), but both charge additional transaction fees of 0.95 euros per transaction and accounting service fees of around 65 euros per hour.

Based on Kluvin Apteekki's current business volume (approximately 630 sales invoices, 840 purchase invoices, and 16 employees' salaries annually), System 3 can avoid the hidden costs of high-frequency trading and be more economical in the long run. As Kluuvin Apteekki expands, System 1 and System 2 fees will increase linearly with transaction volume, making it harder to predict long-term costs. In contrast, System 3 offers more controllable long-term costs and is better suited for business expansion (e.g., opening new branches).

2). Customization

System 3 provides moderate customization upon request, which allows the system to adapt as Pia's business grows and her needs change.

Although System 2 is function-oriented and offers some integrations similar to System 3, its customization options are more limited, which might not fully accommodate Pia's requirement as the business expands. System 1 also provides standard settings with no customization options, which could be restrictive as the business expands.

3). Service Integration

System 3 integrates accounting services directly into the system, simplifying tasks such as payroll calculations, tax reporting and annual reports. This reduces the complexity and time Pia spends coordinating across different systems.

While System 2 provides some third-party integration, it lacks the deep integration with accounting services, which means Pia would need to work with external accounting firms, potentially leading to communication and service inconsistencies. And System 1 focuses more on connecting with the client company and does not integrate accounting services, meaning Pia would also need to coordinate with external accounting providers.

4). Reliability and Compliance

With 30 years of industry experience and being a Finnish system provider, System 3 provides greater data security and system stability and better alignment with Finnish laws and GDPR regulations. This makes it a more secure and compliant solution for Pia's business, which deals

with sensitive payroll and tax data. In addition, its long-term existence also ensures strong support, which is critical as Pia's business grows and faces more complex regulatory requirements.
Although the System 2 provider is also based in Finland, its experience is shorter (14 years) compared to System 3, which could result in less stability in the long run. As for System 1, being an international provider, it may lack the localized compliance support needed to fully align with Finnish regulations, which could pose unnecessary legal risk for Pia. And System 1 has the shortest business year (9 years), which could suggest lower reliability and potential instability.

2. Should Pia be worried about possible lock-in with either accountant or system provider?

Lock-in concerns arise when a company gets dependent on a specific technology or service provider, making transfers costly or difficult. Pia must carefully evaluate the risks associated with the accountant and the accounting system in order to avoid problems in the future.

Reliance on a specific accountant is a severe problem. If Pia chooses an accountant that uses proprietary tools or alters financial procedures, it could be challenging to switch to a new provider. A shift could be challenging due to differences in accounting practices, data architecture, and reporting formats. Contractual agreements may also include restricted provisions or termination costs, which would make transferring much more difficult.

The same is true with cloud-based accounting solutions, which include hazards. Some suppliers utilize proprietary software that limits data portability, making it challenging to transfer financial records to a new system. If the system is incompatible with other business software, Pia may have to continue with the same provider. Furthermore, price structures are subject to change over time, raising the cost of long-term use above early estimates.

Pia may lessen these risks, though, by choosing System 3 as her accounting program. The seamless data export capability of System 3 will make it easy to transfer financial information if she decides to switch providers in the future. Furthermore, the application easily connects with third-party programs, reducing the likelihood of being limited to a single ecosystem. Additionally, because System 3 offers variable service plans that release her from long-term commitments, Pia may adjust her consumption to meet business needs.

To further protect herself from lock-in, Pia should ensure that her contract with the accountant is devoid of restrictive language that would limit her freedom to move service providers. Regularly storing up financial data in an accessible manner might provide additional security.

3. How should Pia mitigate the possible data security risks?

Since financial data is very sensitive and must be protected from breaches, unauthorized access, and data loss; therefore, data security is a crucial consideration when implementing a cloud-based accounting system. Pia must take proactive measures to safeguard the security and integrity of her accounting data.

One of the primary concerns is the potential for unauthorized access. Weak authentication processes might allow hackers to exploit vulnerabilities in the system. Additionally, if the cloud provider has a security breach, Pia's financial information may become public knowledge. Data loss due to system failures, accidental deletions, or cyberattacks is another serious risk that might result in financial and operational interruptions. Following financial and data protection laws is also necessary to prevent legal issues.

System 3 has strong security mechanisms to lower these dangers. It incorporates multi-factor authentication (MFA) to prevent unauthorized access and employs advanced encryption for both stored and transmitted data. The system's ability to allow automatic backups reduces the likelihood of irreversible data loss in the case of faults or cyber attacks. System 3 also offers real-time activity monitoring, which makes it easier to identify and address suspicious activities quickly.

To increase security, Pia should ensure that financial data is only accessible by those who are authorized. System3 user roles may be explicitly defined to prevent undesired modifications and accidental data disclosure. She should also make sure the system complies with all relevant financial rules in order to lower legal risks. By using System3's security features and implementing internal safeguards, Pia may confidently transition to a cloud-based accounting system and secure her financial data from any dangers.

4. Is price an important factor when choosing the system?

Yes, price is an important factor when choosing the system but it should not be the only factor. It is important to balance cost with the cost with the features and value the system provides, especially for small businesses like Kluuvin Apteekki. While a lower price may seem appealing initially, it is important to consider other factors like hidden costs, long-term value, scalability and the level of support provided by the system.
For instance, systems with low upfront costs, like System 1 and System 2, may seem more affordable but may incur transaction fees and require additional accounting services. These costs can accumulate quickly, making them potentially more expensive than a system like System 3, which has a higher upfront cost but provides predictable pricing and bundled accounting services.
Thus, while price is an important consideration, the overall value for money—considering future growth, scalability, and the system's ability to meet the expanding needs of the business—is equally important. System 3 may have a higher initial cost, but it offers greater long-term value,

service integration, and reliability, which ultimately makes it a more cost-effective choice for Pia's business.

5. What is the optimal division of work between Pia and the accountant? Should Pia outsource all the accounting tasks, use selective outsourcing, or keep all tasks to herself?

The best choice for Pia is to **use selective outsourcing** instead of either outsourcing everything or doing all tasks herself. This way, she can have professional help with tasks that are complicated or time-consuming, while still keeping control over important parts of the business.

Here's a breakdown of each task:

- **Sale Tasks:**
  - Outsource: Sending and handling sales invoices and maintaining the sales ledger are repetitive and routine. An accountant can handle these tasks more efficiently.
  - In-House: Maintaining the client and product registers should be done by Pia because these are closely tied to customer relationships and require detailed knowledge of her business.
- **Purchases Tasks:**
  - Outsource: Handling purchase invoices, processing travel/other costs, and maintaining the purchases ledger are standard and can be processed by an external provider.
  - In-House: Supplier register maintenance is important for managing supplier relationships directly, so Pia should keep this task in-house.
- **Payroll:**
  - Outsource: Payroll calculations are complex (especially with part-time employees) and error-prone, so it makes sense to have a professional handle these.
  - In-House: Maintaining employee records (personnel register and basic payroll data) should be done internally because it requires up-to-date, detailed information that Pia knows best.
- **Reporting:**
  - Outsource: The preparation of balance sheets, income statements, VAT, annual salary, and pension insurance reports involves a lot of detailed work and expert knowledge, making them ideal for professional outsourcing.
  - In-House Oversight: Although these reports can be prepared by an accountant, Pia should review them herself to ensure they match her business situation and comply with regulations.
- **Payments:**
  - A hybrid approach will work best:
    Routine tasks like VAT payments, salary disbursements, and other routine payments might be automated or handled by the accountant.
    However, larger or more critical payments should be closely monitored by Pia.

Implications:

- Efficiency: Outsourcing routine tasks frees up Pia's time so she can focus on managing and growing her business.
- Risk Reduction: Professional accountants can reduce errors in complex tasks (like payroll and reporting) that might lead to legal or financial problems.
- Control: By keeping key tasks (like customer and supplier registers) in-house, Pia maintains direct control over her business's critical information.

6. How to ensure a good fit between the selected system and the work arrangement?

To make sure the chosen cloud-based accounting system (System 3) fits well with the way work is divided between Pia and her accountant, several points need to be considered:

1. **Customization and Integration:** The system should allow some customization so that it can be set up to handle both in-house tasks and outsourced ones. For example, it might have separate sections or modules for sales, purchases, payroll, and reporting.
2. **Real-Time Monitoring:** The cloud system should have a dashboard or a way for Pia to see real-time updates. This way, even if tasks are outsourced, she can quickly check on their progress and accuracy.
3. **Scalability:** As the pharmacy grows, the system should be able to handle more transactions and possibly new types of accounting tasks. It should be flexible enough to adapt to changes in the business.
4. **Clear Service Level Agreements (SLAs):** There should be clear SLAs with the external accountant. These agreements would set out expectations for turnaround times, quality of work, and how issues are resolved.
5. **Data Security and Compliance:** Since the system will hold sensitive financial and payroll data, it must meet high security standards (like those required by GDPR) to protect Pia's business information.
6. **Training and Support:** Both Pia's staff and the external accountants should receive proper training on how to use the system. Additionally, ongoing support from the system provider will help address any issues quickly.

By making sure the cloud system supports these features, Pia can have an efficient, secure, and adaptable setup that matches the selective outsourcing strategy.

7. What are the transaction costs incurred by the decision?

Initially, transaction costs manifest as search and information expenses, which encompasses the considerable time and effort Pia dedicates to evaluating various cloud solutions, assessing vendor reliability, regulatory compliance, scalability, and integration possibilities. Potential advisory fees from external consultants also add to these initial expenditures.

Subsequently, negotiation and contracting costs emerge, including legal consultations and time investments required to finalize contracts. Once the contracting phase concludes, Pia faces substantial implementation and integration costs, which involves adapting existing business processes, providing extensive staff training, integrating new software with existing systems (such as banking and tax interfaces), and securely migrating historical accounting data into the new AIS.

After the stage of implementation, Pia continues to incur transaction costs associated with ongoing monitoring and coordination efforts to ensure that the AIS operates within agreed-upon standards. Regular audits, quality assurance, and compliance checks form essential parts of this continuous oversight process.

8. What about long term strategy? Reflect on the implications of your choices in the future.

Selecting System 3—a hybrid cloud-based AIS combined with outsourced accounting services—strategically aligns with Pia's long-term business goals, operational efficiency, and predictable cost management. However, this decision comes with opportunities and strategic risks, shaping future operations.

Opportunities and Benefits

- Scalability and Flexibility: Enables Pia's pharmacy to easily handle increased transaction volumes, adapt to regulatory changes, and swiftly respond to market conditions without significant disruptions.
- Integration Capability: Smooth integration with existing tools (banking systems, tax reporting, CRM solutions) allows efficient operation and quick adaptation to new business needs.
- Focus on Core Business: Outsourcing routine accounting tasks allows Pia to concentrate strategically on pharmacy operations, customer relations, and innovation, enhancing competitive advantage.
- Predictable Cost Structure: Stable and predictable costs facilitate accurate financial forecasting, improved cash flow management, and confident investment in growth initiatives.
- Expert Regulatory Compliance: Professional accountants ensure ongoing adherence to evolving local regulations, significantly reducing compliance risks and potential reputational damage.

Strategic Risks and Considerations

- Vendor Lock-in and Dependency: Reliance on external providers increases risks related to vendor lock-in, where switching providers might incur high costs or disruptions. Maintaining contract flexibility and data portability mitigates these risks.

- Data Security and Privacy: Increased reliance on cloud services and outsourcing introduces potential cybersecurity threats and data privacy concerns. Strong contractual safeguards, regular security audits, and proactive cybersecurity measures are crucial.
- Erosion of Internal Expertise: Outsourcing might gradually diminish internal accounting knowledge, increasing dependence on external advisors. Regular training and retaining key accounting functions internally help preserve strategic independence.
- Operational Continuity Risk: Potential operational interruptions due to provider downtime, technical failures, or communication issues could impact daily operations. Strong contractual guarantees and contingency planning can reduce these risks.

Strategic Recommendations and Future Actions

To maximize long-term benefits and mitigate risks, Pia should:

- Regularly evaluate vendor performance, data security, and compliance measures.
- Establish clear contracts emphasizing service flexibility, data portability, and security obligations.
- Maintain internal accounting capabilities through continuous staff training.
- Implement strong internal control frameworks to ensure outsourced operations meet quality and compliance standards.

To sum up, choosing System 3 offers significant long-term strategic benefits if risks like scalability, integration, vendor lock-in, and cybersecurity are actively managed. By addressing these considerations through strategic planning and strong contractual frameworks, Pia's pharmacy will sustain operational efficiency and competitive advantage well into the future.